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MATHEMATICS 2210
Quiz 3, Fall semester 2012

Name:....KEY....

5 pts **1.** Let $\bar{r}(t) = (\ln(t^2), t^2, t^3)$. Compute $\bar{r}'(t) \times \bar{r}''(t)$.

$\bar{r}' = \left(\frac{2\cancel{t}}{t^{\cancel{2}}}, 2t, 3t^2 \right) \qquad \bar{r}'' = \left(\frac{-2}{t^2}, 2, 6t \right)$

$$\bar{r}' \times \bar{r}'' = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2/t & 2t & 3t^2 \\ \frac{-2}{t^2} & 2 & 6t \end{vmatrix} = \vec{i} \begin{vmatrix} 2t & 3t^2 \\ 2 & 6t \end{vmatrix} - \vec{j} \begin{vmatrix} \frac{2}{t} & 3t^2 \\ \frac{-2}{t^2} & 6t \end{vmatrix} + \vec{k} \begin{vmatrix} \frac{2}{t} & 2t \\ \frac{-2}{t^2} & 2 \end{vmatrix}$$

$$= \left(6t^2, -18, \frac{8}{t} \right)$$

5 pts **2.** Let $\bar{r}(t) = (t, t^2, t^3)$.
a) Write the tangent line to $\bar{r}(t)$ at $t = 1$ (in parametric form).
b) Write the plane perpendicular to $\bar{r}(t)$ at $t = 1$.

$\bar{r}' = (1, 2t, 3t^2) \qquad \bar{r}'(1) = (1, 2, 3) \qquad \bar{r}(1) = (1, 1, 1)$

a) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1+t \\ 1+2t \\ 1+3t \end{pmatrix}$

b) $x + 2y + 3z = D \qquad 1 + 2 + 3 = D$

$x + 2y + 3z = 6$